

# Onboard SAR Processing for Ship Detection: Modeling and Evaluation

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# Outline

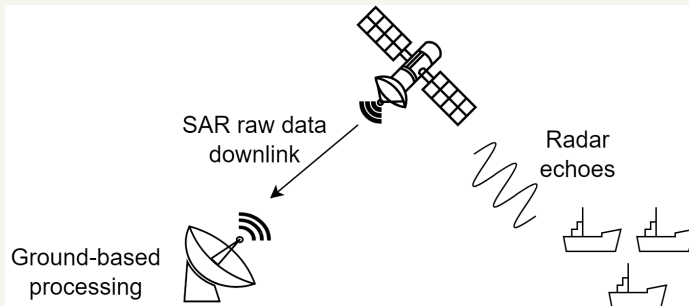
- ▶ Introduction and Motivation
- ▶ Evaluation Framework
- ▶ SAR Echo and Imaging Model
- ▶ Sparse Image Refinement
- ▶ Experimental Results
- ▶ Conclusion
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# Introduction

- ▶ SAR raw data are downlinked for ground-based processing, including SAR focusing, image enhancement, and ship detection.
- ▶ Detected ships provide information for maritime surveillance.



# Problem and Motivation

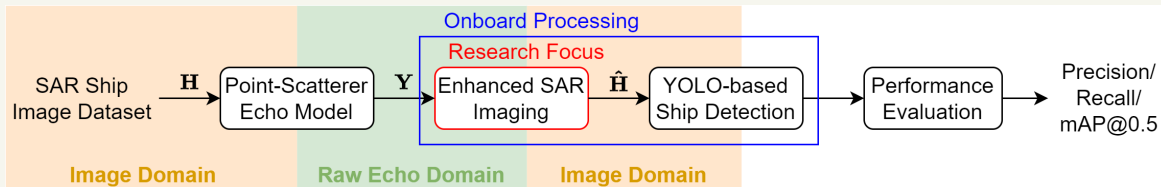
- ▶ Ground-Based Processing
  - ▶ Large raw-data volume
  - ▶ Limited downlink bandwidth
  - ▶ Delayed target information
- ▶ Onboard Processing
  - ▶ Reduced data transmission and detection latency
  - ▶ Challenge: Limited computation, memory, and power
- ▶ Research Goal
  - ▶ Lightweight SAR processing for low-latency ship detection

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# Evaluation Framework

- Research Focus: Enhanced SAR imaging
- Evaluation Task: YOLO-based ship detection



## ► Evaluation Metrics [1]

- Precision:  $\frac{TP}{TP+FP}$ , Recall:  $\frac{TP}{TP+FN}$

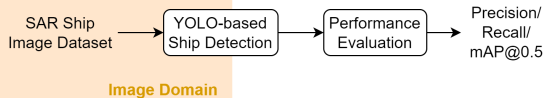
TP: True Positive, FP: False Positive, FN: False Negative

- mAP@0.5: Mean AP across classes at IoU threshold = 0.5

IoU = overlap area / union area of predicted and ground-truth boxes

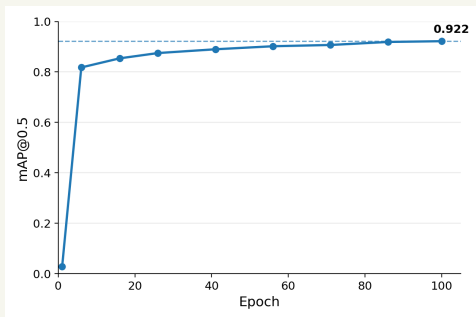
# Dataset and Detection Model Setup

- ▶ HRSID [2]
  - ▶ High-Resolution SAR Images Dataset
  - ▶ Training: 3,642 images
  - ▶ Validation: 1,962 images
  - ▶ Image size:  $800 \times 800$  pixels
- ▶ YOLO Detection Model
  - ▶ YOLO11m fine-tuned on HRSID (100 epochs)



## Validation Performance

- ▶ Precision: 0.910
- ▶ Recall: 0.840
- ▶ mAP@0.5: 0.922



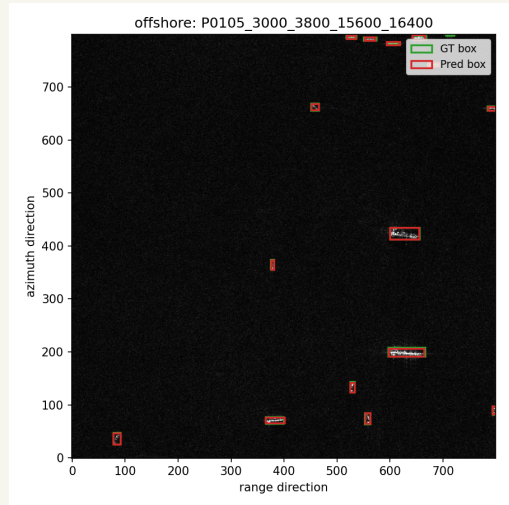


# YOLO Ship Detection Results (1)

|           | <b>Overall</b> | <b>Inshore</b> | <b>Offshore</b> |
|-----------|----------------|----------------|-----------------|
| Images    | 1,962          | 369            | 1,593           |
| Precision | 0.913          | 0.803          | 0.982           |
| Recall    | 0.846          | 0.741          | 0.965           |
| mAP@0.5   | 0.918          | 0.817          | 0.984           |

- Offshore scenes achieve better detection performance than inshore scenes [3].

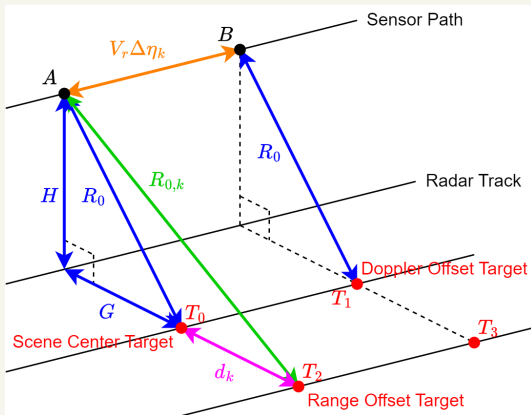
# YOLO Ship Detection Results (2)



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# Point-Scatterer Echo Model: Geometry



- Target position determines the time-varying slant range  $R_k(\eta_m)$ .
- $$R_k(\eta) = \sqrt{R_{0,k}^2 + (\eta_m + \Delta \eta_k)^2 V_r^2}$$
  - $R_{0,k} \approx R_0 + d_k$
  - $R_0$ : slant range to scene center
  - $d_k$ : ground-range offset of scatterer  $k$ , approximated as slant-range offset
  - $\eta_m = m \Delta \eta$ : slow time
  - $\Delta \eta_k$ : azimuth-time offset of scatterer  $T_k$

# Point-Scatterer Echo Model: Signal Synthesis

## ► Geometry-Dependent Delay and Phase

$$\text{► } \tau_k(\eta_m) = \frac{2R_k(\eta_m)}{c}, \phi_k(\eta_m) = -4\pi \frac{R_k(\eta_m)}{\lambda}$$

## ► Echo Synthesis: $\mathbf{S} = [s_{m,n}]_{M \times N}$

$$s_{m,n} = \sum_k A_k \text{rect}\left(\frac{\tau_n - \tau_k(\eta_m)}{T_r}\right) \cdot e^{j\pi K_r (\tau_n - \tau_k(\eta_m))^2} \cdot e^{j\phi_k(\eta_m)}$$

►  $\eta_m = m\Delta\eta$ : slow time,  $\tau_n = n\Delta\tau$ : fast time

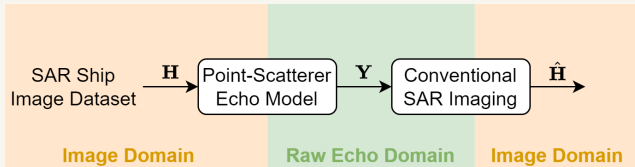
►  $A_k$ : scattering coefficient of scatterer  $k$ ,  $K_r = \frac{B_r}{T_r}$

$K_r$ : chirp rate,  $B_r$ : chirp bandwidth (Hz),  $T_r$ : pulse duration (sec)

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# Conventional SAR Imaging



- Imaging goal: Recover  $\mathbf{H}$  from  $\mathbf{Y}$ .
- Conventional methods: RDA, CSA, and BP

RDA: Range Doppler Algorithm, CSA: Chirp Scaling Algorithm, BP: Back-Projection

# One-Step Sparse Image Refinement

- Image-domain regularization [4]

$$\hat{\mathbf{H}} = \arg \min_{\mathbf{H}} \underbrace{\frac{1}{2} \|\mathbf{H} - \mathbf{H}_{\text{CSA}}\|_F^2}_{\text{Preserve CSA image}} + \lambda \underbrace{\|\mathbf{H}\|_1}_{\text{Promote sparsity}}$$

- Strong scatterers  $\rightarrow$  retained; weak components  $\rightarrow$  suppressed
- $\hat{\mathbf{H}} = \mathcal{S}_{\lambda}(\mathbf{H}_{\text{CSA}})$ ,  $\mathcal{S}_{\lambda}(x) = \frac{x}{|x|} \max(|x| - \lambda, 0)$ ,  $x \neq 0$
- One-Step Refinement
  - $\mathbf{Y} \xrightarrow{\text{CSA}} \mathbf{H}_{\text{CSA}} \xrightarrow{\mathcal{S}_{\lambda}(\cdot)} \hat{\mathbf{H}}$



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# Simulation Setup

- ▶ Dataset: 100 offshore SAR images sampled from 1,593 images
- ▶ Evaluation: Ship detection on reconstructed SAR images
- ▶ SAR Simulation Parameters

| Parameter                  | Value        | Parameter           | Value    |
|----------------------------|--------------|---------------------|----------|
| Pulse Width                | 12.5 $\mu$ s | Wavelength          | 0.1152 m |
| Pulse Repetition Frequency | 1 kHz        | Closest Slant Range | 4000 m   |
| Bandwidth                  | 50 MHz       | Platform Velocity   | 120 m/s  |
| Sampling Frequency         | 64 MHz       | Antenna Length      | 1.2 m    |

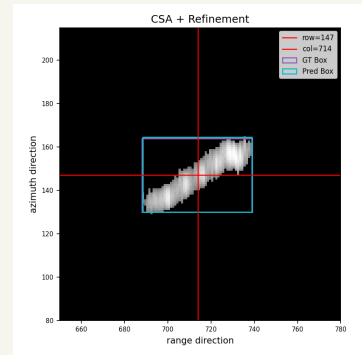
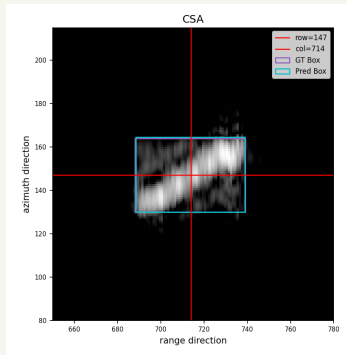
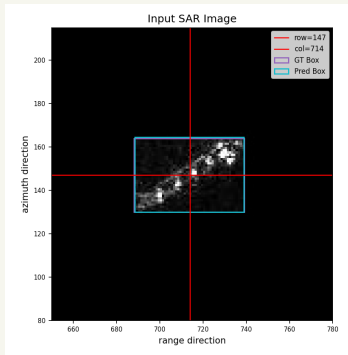
# Ship Detection Performance

| Image            | Precision | Recall | mAP@0.5 |
|------------------|-----------|--------|---------|
| Input SAR Image  | 1.000     | 0.978  | 0.975   |
| CSA              | 0.977     | 0.940  | 0.954   |
| CSA + Refinement | 0.993     | 0.986  | 0.985   |

- Refinement threshold set to 120 (intensity range: 0–255)
- Thresholding suppresses CSA-induced background artifacts and improves ship detection performance.

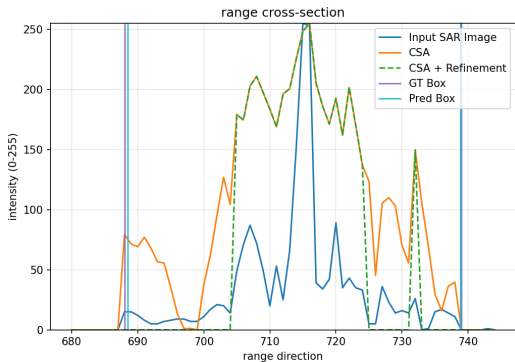
# Representative Example

- Observation: Thresholding suppresses background artifacts while preserving the dominant ship response.



# Cross-Section Analysis

- Thresholding removes low-amplitude sidelobe/background responses while retaining the dominant target structure.



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# Conclusion

- ▶ Established an end-to-end framework for evaluating SAR image enhancement through ship detection.
- ▶ One-step sparse refinement suppresses background artifacts while preserving dominant ship responses.
- ▶ CSA + refinement improves detection performance from 0.954 to 0.985 mAP@0.5.

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# Reference

- [1] F. Song, S. Yang and X. Zhou, “SARES-DEIM: Sparse Mixture-of-Experts Meets DETR for Robust SAR Ship Detection,” in *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing (JSTARS)*, 2026.
- [2] S. Wei, X. Zeng, Q. Qu, M. Wang, H. Su and J. Shi, “HRSID: A High-Resolution SAR Images Dataset for Ship Detection and Instance Segmentation,” in *IEEE Access*, vol. 8, pp. 120234–120254, 2020.
- [3] T. Zhang *et al.*, “Balance Scene Learning Mechanism for Offshore and Inshore Ship Detection in SAR Images,” in *IEEE Geoscience and Remote Sensing Letters*, vol. 19, pp. 1–5, 2022.
- [4] H. Bi, G. Bi, B. Zhang, W. Hong and Y. Wu, “From Theory to Application: Real-Time Sparse SAR Imaging,” in *IEEE Transactions on Geoscience and Remote Sensing*, vol. 58, no. 4, pp. 2928–2936, April 2020.